

## Question

At  $-78^{\circ}\text{C}$ ,  $\text{Cl}_2$ ,  $\text{NCl}_3$ , and  $\text{AsF}_5$  react in liquid  $\text{SO}_2$  solvent to produce a salt **A**. Both the cation and anion of **A** exhibit high symmetry, and the molar ratio of  $\text{Cl}_2$  to  $\text{NCl}_3$  in the reaction is 1:2. Upon heating, **A** decomposes without yielding any elemental substances. When **A** reacts with  $\text{CF}_3\text{SCl}$ , it forms salt **B** and several small-molecule binary compounds, with the original  $\text{S}-\text{Cl}$  bond remaining intact. The symmetry of the cation in **B** is identical to that of  $\text{NH}_3$ .

Regarding the above process, which of the following statements is incorrect?

- A.** The anions of **A** and **B** are identical, and both belong to the  $\text{O}_h$  point group.
- B.** The cation of **A** belongs to the  $\text{T}_d$  point group.
- C.** During the generation of **A**, the product of the simplest integer ratio of the balanced chemical equation coefficients is 12.
- D.** The gas produced during the decomposition of **A** corrodes glass containers in a moist state, so it is necessary to ensure the experiment is strictly anhydrous.
- E.** During the generation of **B**, the cations in **B** and one of the products are isoelectronic.
- F.** During the generation of **B**, the sum of the simplest integer ratio of coefficients in the balanced chemical equation is 8.
- G.** All of the above statements are correct
- H.** All other statements are incorrect

## Answer

Correct Answer: **G**

## Detailed Explanation

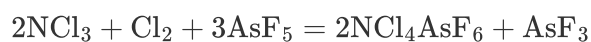
This is an inorganic compound inference problem. The question states that "both the cation and anion of **A** possess high symmetry." Considering that the only reactant capable of forming the anion is  $\text{AsF}_5$ , the anion should be  $\text{AsF}_6^-$ . Since a reaction must occur, the cation should not contain As. Therefore, the reactant involved in forming the cation is  $\text{NCl}_3$ , and the resulting cation should be  $\text{NCl}_4^+$ .

### CHECKPOINT

1 PTS

Compound **A** is  $\text{NCl}_4\text{AsF}_6$

Balancing the equation based on the hint "the molar ratio of  $\text{Cl}_2$  to  $\text{NCl}_3$  is 1:2," the chemical equation for this process is:



### CHECKPOINT

1 PTS

The chemical equation for the formation of **A** is  $2\text{NCl}_3 + \text{Cl}_2 + 3\text{AsF}_5 = 2\text{NCl}_4\text{AsF}_6 + \text{AsF}_3$

According to the question, "when heated, **A** decomposes without producing elemental substances," indicating that the decomposition products cannot include  $\text{Cl}_2$  or  $\text{N}_2$ . We can deduce the fate of the elements: during heating, an oxidizing gas must form, and the reduction of As is impossible, so As exists as  $\text{AsF}_5$  in the products. Since  $\text{N}_2$  is absent, there is no stable high oxidation state of N. The oxidation of N cannot occur during heating, so N

remains as  $\text{NCl}_3$ . Considering element conservation and the absence of elemental substances in the products, the product should be  $\text{ClF}$ . The balanced chemical equation is:


**CHECKPOINT**

1 PTS

The decomposition equation of **A** is  $\text{NCl}_4\text{AsF}_6 = \text{NCl}_3 + \text{ClF} + \text{AsF}_5$

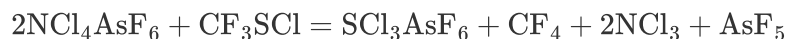
$\text{NCl}_4\text{AsF}_6$  reacts with  $\text{CF}_3\text{SCl}$ . Based on the question's statement that "the original  $\text{S}-\text{Cl}$  bond remains intact," it can be inferred that the  $\text{C}-\text{S}$  bond breaks during the reaction. According to the hint "the symmetry of the cation in **B** is the same as that of  $\text{NH}_3$ ," we know **B** is also composed of cations and anions. Observing the reactants capable of forming cations, the cation can only be formed from  $\text{S}$ ,  $\text{Cl}$ , and  $\text{CF}_3$ . Considering the symmetry matches  $\text{NH}_3$ , the cation should be  $\text{SCl}_3^+$ . Examining the reactants capable of forming anions, it is clear that the anion is solely  $\text{AsF}_6^-$ .

**CHECKPOINT**

1 PTS

Compound **B** is  $\text{SCl}_3\text{AsF}_6$

Considering element conservation and the formation of small binary molecular compounds, the products are  $\text{CF}_4$  and  $\text{NCl}_3$ . The balanced equation is:



## CHECKPOINT

1 PTS

The chemical equation for the formation of **B** is

$$2\text{NCl}_4\text{AsF}_6 + \text{CF}_3\text{SCl} = \text{SCl}_3\text{AsF}_6 + \text{CF}_4 + 2\text{NCl}_3 + \text{AsF}_5$$

Analyzing the options:

The anions of both **A** and **B** are  $\text{AsF}_6^-$ , belonging to the  $\text{O}_h$  point group. Option A is correct.

## CHECKPOINT

1 PTS

The anions of both **A** and **B** are  $\text{AsF}_6^-$ , belonging to the  $\text{O}_h$  point group. Option A is correct.

The cation of **A** is  $\text{NCl}_4^+$ , belonging to the  $\text{T}_d$  point group. Option B is correct.

## CHECKPOINT

1 PTS

The cation of **A** is  $\text{NCl}_4^+$ , belonging to the  $\text{T}_d$  point group. Option B is correct.

In the formation of **A**, the product of the simplest integer ratio of the balanced equation coefficients is  $2 \times 2 \times 3 = 12$ . Option C is correct.

## CHECKPOINT

0.5 PTS

In the formation of **A**, the product of the simplest integer ratio of the balanced equation coefficients is  $2 \times 2 \times 3 = 12$ . Option C is correct.

The decomposition of **A** produces  $\text{ClF}$ , which reacts with water to form  $\text{HF}$ , capable of etching glass. Option D is correct.

## CHECKPOINT

1 PTS

The decomposition of **A** produces  $\text{ClF}$ , which reacts with water to form  $\text{HF}$ , capable of etching glass. Option D is correct.

In the formation of **B**,  $\text{SCl}_3^+$  and  $\text{NCl}_3$  are isoelectronic. Option E is correct.

## CHECKPOINT

1 PTS

In the formation of **B**,  $\text{SCl}_3^+$  and  $\text{NCl}_3$  are isoelectronic. Option E is correct.

In the formation of **B**, the sum of the simplest integer ratio of the balanced equation coefficients is  $2 + 1 + 1 + 1 + 2 + 1 = 8$ . Option F is correct.

## CHECKPOINT

1 PTS

In the formation of **B**, the sum of the simplest integer ratio of the balanced equation coefficients is  $2 + 1 + 1 + 1 + 2 + 1 = 8$ . Option F is correct.

All the above statements are correct, so the answer is G.